

Don't guess where to open. Ask the map.

An AI-assisted WebGIS for retail and F&B site selection around Jakarta's transit network. Demand, competition and available space, scored per walkable cell and explained in plain language.

Team Triple T
Universitas Bina Nusantara
MAPID WebGIS Competition 2026
Maps That Think! · Final

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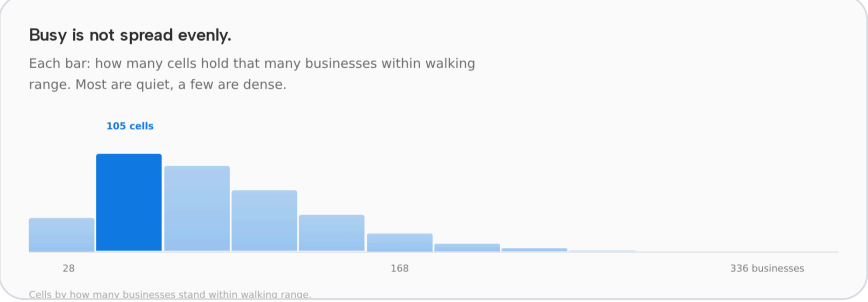


A street block modelled from real counts. The outlined plot is the business you could open.

2 Background & Problem Statement

The blocks around Jakarta's MRT, KRL, LRT and TransJakarta stops are the busiest trading ground in the city. Yet most owners still choose a location on instinct, a hunch, or by copying a competitor. The wrong choice means an early closure and capital lost.

- 1 Nobody counted how busy it is**
Everyone knows station blocks are busy. Nobody knows which blocks.
- 2 The competition is invisible**
A street already full of coffee shops is a way to lose money.
- 3 The space is never counted**
A good area with nothing to rent is not an opportunity.



Busy is not spread evenly. Cells by how many businesses stand within walking range: most are quiet, a few are dense.

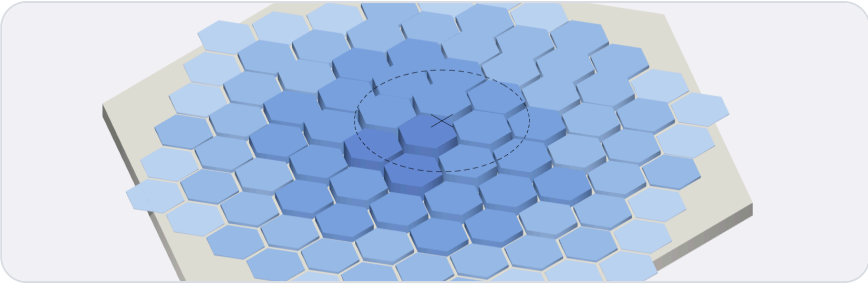
3 Objective & Solution

OBJECTIVES

- Bring demand, competition and available space into one Opportunity Score per cell and business type.
- Explain every score step by step, with the number of data points behind it.
- Say "not surveyed" where data is missing, instead of interpolating a plausible value.

SOLUTION

SpotOn is a decision-support WebGIS. Jakarta is covered with an H3 hexagonal grid, and the **562 cells** with a transit stop within an 800 m walk are scored for **13 business types**. Tapak, the AI guide inside the map, answers plain-language questions in Indonesian or English with the reasons attached.



One hexagon, one cell. Height and colour both carry the Opportunity Score, sampled from the grid's real scores.

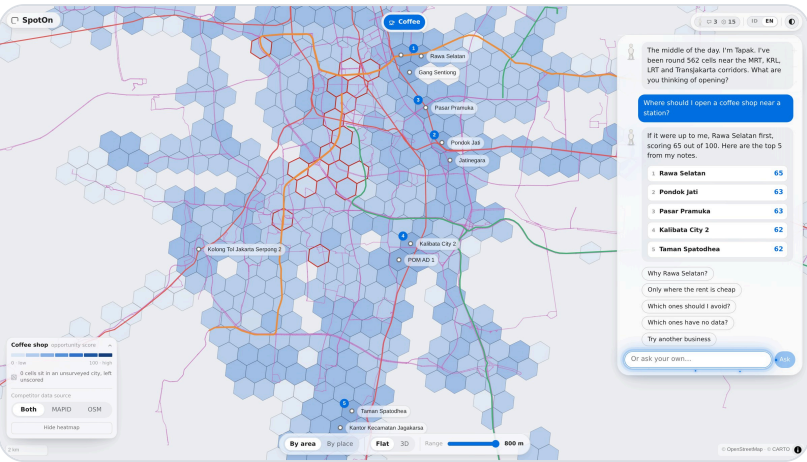
4 Data & Methodology

SOURCE	WHAT IT CONTRIBUTES	COUNT
MAPID Data Premium	Competitor points, 13 types	24,630
MAPID Apps Missions	Field records, 4 missions	1,027
OSM Overpass	Transit stops, POIs, hours	1,105
MAPID Property	Listings with asking prices	3,547

- 1 H3 grid**
Resolution 8, 800 m walk, transit-served cells only.
- 2 Spatial join**
Rivals, trade, stops, listings and notes per cell.
- 3 Score and rank**
 $\text{clamp}(\text{busy} - \text{rivals}) \times \text{space gate} \times \text{access} \times \text{cost}$, per type.
- 4 AI layer**
The model picks the operation. The engine computes every figure.

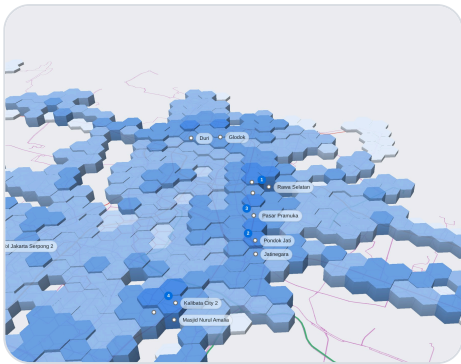
Field records never enter the score. A grounding check drops any AI sentence quoting a figure the engine did not compute.

5 WebGIS & Key Features

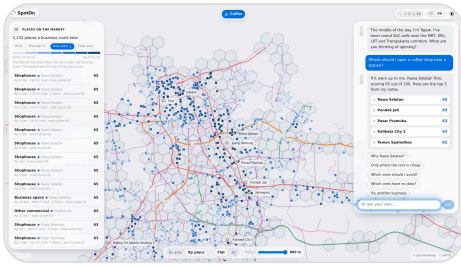


Ask "Where should I open a coffee shop near a station?" and the map recolours by Opportunity Score, ranks the cells and lists the reasons.

- A Interactive map on MAPID MAPS**
562 hexagons, flat or 3D. Zoom, click, filter, layer control, attribute table, ID and EN.
- B Tapak, the AI guide**
Ask in plain language. The map recolours and flies to the answer. Follow up with "why that one".
- C Area card**
Score breakdown, competitors, transit nodes, opening hours, cost of space and field notes.
- D Place pivot**
Every premise on the market, ranked with the score of its area. Nobody rents a hexagon.
- E Your own weights**
Demand against competition, walking radius, space requirement, competitor source.



3D view: height carries the same figure as colour.



By place: 2,232 premises on the market, sortable by price, area score or floor area.

How a reader interacts. Type a question or tap a suggestion. The map answers. Click a cell or a result and the card explains it. Ask a follow-up in the same thread. Every figure on screen is computed by the scoring engine from the data, and the model only writes the sentence around it.

6 Results & Insights

- 462**
cells scored, with the N behind every figure
- 100**
cells left unscored: no source has surveyed their city yet

COFFEE SHOPS, ALL 562 CELLS

37 underserved 508 competitive 17 saturated

TOP CATCHMENTS	SCORE	RIVALS	UNITS
Rawa Selatan	65	5	12
Pondok Jati	63	10	9
Pasar Pramuka	63	14	5

Why Rawa Selatan? 273 other businesses within 800 m put the trade around it at 96 out of 100. Only 5 coffee shops stand in the same range, so competition reads 9. 12 units are on the market at a median of Rp 40.2m per m², and 4 transit nodes are within walking range.

Minimarkets tell the opposite story: 529 cells competitive, 32 saturated, only 1 underserved. The gaps are few and specific, which is what a site finder needs.

709 of the 1,027 field records fall inside the grid, across 191 cells. 55 property records offer space for rent, the only rental signal for Jakarta.

7 Benefits & Applications

BENEFITS

- Less risk of opening in the wrong place.
- Small businesses compete on data, with no GIS skills.
- Every figure traceable to its source and its N.

TARGET USERS AND APPLICATIONS

- Investors and franchise teams** screening transit sites.
- First-time owners** asking what fits near home.
- Survey planning:** unscored cells guide the next mission.
- Other cities:** swap the transit data and catalogue.

8 Conclusion

Site selection around transit was decided on instinct because demand, competition and available space were never counted together. SpotOn counts them per walkable cell from MAPID and OpenStreetMap data and turns them into one transparent Opportunity Score for 13 business types. The result is insight rather than raw points: ranked catchments, a typology for every cell, a card that takes each score apart, and an AI guide that explains it while every number stays computed by the engine.

FUTURE DEVELOPMENT

- Grow the field surveys to cover the 100 unscored cells.
- Read formality tier and storefront quality from mission photos.
- Rent listings, more business types, a mobile-first card.

9 Open the WebGIS



spot-on-three.vercel.app

Scan to open SpotOn. Desktop and mobile, Indonesian and English, no install.

Source: github.com/VennethN/SpotOn. SvelteKit and MapLibre GL on the MAPID MAPS basemap, deployed on Vercel.

Transit and business data © OpenStreetMap contributors (ODbL). Business points, listings and field notes from MAPID Data Premium and MAPID Apps, within the competition's terms.